EXPERIMENT-17

Implement Mobile Price Prediction using a suitable Machine Learning algorithm.

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
print("=" * 70)
print("            MOBILE PRICE PREDICTION")
print("=" * 70)

# ------------------------------------------------------
# STEP 1: UPLOAD DATASET
# ------------------------------------------------------
uploaded = files.upload()
filename = list(uploaded.keys())[0]
data = pd.read_csv(filename)
print("\nDataset Loaded Successfully!")

# ------------------------------------------------------
# STEP 2: DISPLAY DATASET
# ------------------------------------------------------
print("\n" + "=" * 70)
print("                MOBILE DATASET")
print("=" * 70)
print(data.head(15))
```

```
print("\nTotal Number of Records :", len(data))
print("\nColumn Names:")
print(list(data.columns))

# ------------------------------------------------------
# STEP 3: FIND TARGET COLUMN
# ------------------------------------------------------
target = None
# Check common target column names
possible_targets = [
    "price_range",
    "Price_Range",
    "price range",
    "Price Range",
    "price",
    "Price",
    "label",
    "Label",
    "target",
    "Target",
    "category",
    "Category"
]
for col in possible_targets:
    if col in data.columns:
        target = col
        break
# If none of the common names exist,
# use the LAST COLUMN as target
if target is None:
```

```
    target = data.columns[-1]
print("\n" + "=" * 70)
print("                    TARGET INFORMATION")
print("=" * 70)
print("\nTarget Column :", target)
print("\nTarget Values:")
print(data[target].value_counts())

# ------------------------------------------------------------
# STEP 4: SELECT NUMERICAL FEATURES
# ------------------------------------------------------------
X = data.drop(columns=[target])
# Keep only numerical columns
X = X.select_dtypes(include="number")
y = data[target]
print("\nNumber of Input Features :", X.shape[1])
print("\nInput Features:")
print(list(X.columns))

# ------------------------------------------------------------
# STEP 5: TRAIN TEST SPLIT
# ------------------------------------------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=42
)
print("\n" + "=" * 70)
print("                    DATA SPLITTING")
```

```
print("=" * 70)
print("\nTraining Records :", len(X_train))
print("Testing Records  :", len(X_test))

# ------------------------------------------------------------
# STEP 6: CREATE MODEL
# ------------------------------------------------------------
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
print("\n" + "=" * 70)
print("              TRAINING RANDOM FOREST")
print("=" * 70)
model.fit(X_train, y_train)
print("\nModel Training Completed Successfully!")

# ------------------------------------------------------------
# STEP 7: PREDICTION
# ------------------------------------------------------------
prediction = model.predict(X_test)
print("\n" + "=" * 70)
print("               ACTUAL VS PREDICTED")
print("=" * 70)
# Display first 30 results
for i in range(min(30, len(y_test))):
    print(
        "Record", i + 1,
        " | Actual Price Category =",
        y_test.iloc[i],
```

```
        " | Predicted Price Category =",
        prediction[i]
    )

# ----------------------------------------------------------
# STEP 8: ACCURACY
# ----------------------------------------------------------
accuracy = accuracy_score(y_test, prediction)
print("\n" + "=" * 70)
print("                MODEL PERFORMANCE")
print("=" * 70)
print("\nAccuracy :", round(accuracy * 100, 2), "%")

# ----------------------------------------------------------
# STEP 9: CONFUSION MATRIX
# ----------------------------------------------------------
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, prediction))

# ----------------------------------------------------------
# STEP 10: FEATURE IMPORTANCE
# ----------------------------------------------------------
print("\n" + "=" * 70)
print("                FEATURE IMPORTANCE")
print("=" * 70)
importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.feature_importances_
})
importance = importance.sort_values(
```

```
    by="Importance",
    ascending=False
)
print(importance.to_string(index=False))


# ------------------------------------------------------------
# FINAL RESULT
# ------------------------------------------------------------


print("\n" + "=" * 70)
print("          MOBILE PRICE PREDICTION COMPLETED")
print("=" * 70)
print("\nTarget Column Used :", target)
print("Number of Features :", X.shape[1])
print("Training Samples   :", len(X_train))
print("Testing Samples    :", len(X_test))
print("Model Used         : Random Forest Classifier")
print("Accuracy           :", round(accuracy * 100, 2), "%")
print("\nFinal Result:")
print("The Random Forest algorithm successfully classified")
print("the mobile phones into their respective price categories.")
print("=" * 70)
```

Sample Input:

```
======================================================================
                MOBILE PRICE PREDICTION
======================================================================
```

Choose Files Mobile.csv

Mobile.csv(text/csv) - 252087 bytes, last modified: 8/10/2026 - 100% done

```
Saving Mobile.csv to Mobile.csv
```

Sample Output:

```
Dataset Loaded Successfully!
...
======================================================================
                    MOBILE DATASET
======================================================================
   Battery_power_mAh Bluetooh  Speed_of_microprocessor Dual_sim Front_camera  \
0           842 mAh       No                      2.2       No     1 pixels
1          1021 mAh      Yes                      0.5      Yes     0 pixels
2           563 mAh      Yes                      0.5      Yes     2 pixels
3           615 mAh      Yes                      2.5       No     0 pixels
4          1821 mAh      Yes                      1.2       No    13 pixels
5          1859 mAh       No                      0.5      Yes     3 pixels
6          1821 mAh       No                      1.7       No     4 pixels
7          1954 mAh       No                      0.5      Yes     0 pixels
8          1445 mAh      Yes                      0.5       No     0 pixels
9           509 mAh      Yes                      0.6      Yes     2 pixels
10          769 mAh      Yes                      2.9      Yes     0 pixels
11         1520 mAh      Yes                      2.2       No     5 pixels
12         1815 mAh       No                      2.8       No     2 pixels
13          803 mAh      Yes                      2.1       No     7 pixels
14         1866 mAh       No                      0.5       No    13 pixels

     4G Internal_memeory_gb Mobile_depth Mobile_weight  Cores_of_processor  \
0    No                7 gb       0.6 cm         188 g                   2
1   Yes               53 gb       0.7 cm         136 g                   3
2   Yes               41 gb       0.9 cm         145 g                   5
3    No               10 gb       0.8 cm         131 g                   6
4   Yes               44 gb       0.6 cm         141 g                   2
5    No               22 gb       0.7 cm         164 g                   1
6   Yes               10 gb       0.8 cm         139 g                   8
7    No               24 gb       0.8 cm         187 g                   4
8    No               53 gb       0.7 cm         174 g                   7
9   Yes                9 gb       0.1 cm          93 g                   5
10   No                9 gb       0.1 cm         182 g                   5
11  Yes               33 gb       0.5 cm         177 g                   8
12   No               33 gb       0.6 cm         159 g                   4
13   No               17 gb       1.0 cm         198 g                   4
14  Yes               52 gb       0.7 cm         185 g                   1


    ...  px_height Pixel_width   Ram_mb Screen_height Screen_weight talk_time  \
... 0   ...    20 ppcm    756 ppcm  2549 mb         9 cm          7 cm         19
    1   ...   905 ppcm   1988 ppcm  2631 mb        17 cm          3 cm          7
    2   ...  1263 ppcm   1716 ppcm  2603 mb        11 cm          2 cm          9
    3   ...  1216 ppcm   1786 ppcm  2769 mb        16 cm          8 cm         11
    4   ...  1208 ppcm   1212 ppcm  1411 mb         8 cm          2 cm         15
    5   ...  1004 ppcm   1654 ppcm  1067 mb        17 cm          1 cm         10
    6   ...   381 ppcm   1018 ppcm  3220 mb        13 cm          8 cm         18
    7   ...   512 ppcm   1149 ppcm   700 mb        16 cm          3 cm          5
    8   ...   386 ppcm    836 ppcm  1099 mb        17 cm          1 cm         20
    9   ...  1137 ppcm   1224 ppcm   513 mb        19 cm         10 cm         12
    10  ...   248 ppcm    874 ppcm  3946 mb         5 cm          2 cm          7
    11  ...   151 ppcm   1005 ppcm  3826 mb        14 cm          9 cm         13
    12  ...   607 ppcm    748 ppcm  1482 mb        18 cm          0 cm          2
    13  ...   344 ppcm   1440 ppcm  2680 mb         7 cm          1 cm          4
    14  ...   356 ppcm    563 ppcm   373 mb        14 cm          9 cm          3

     3G touch_screen wifi     price_range
    0    No           No  Yes     Medium cost
    1   Yes          Yes   No       High cost
    2   Yes          Yes   No       High cost
    3   Yes           No   No       High cost
    4   Yes          Yes   No     Medium cost
    5   Yes           No   No     Medium cost
    6   Yes           No  Yes  Very High cost
    7   Yes          Yes  Yes        Low cost
    8   Yes           No   No        Low cost
    9   Yes           No   No        Low cost
    10   No           No   No  Very High cost
    11  Yes          Yes  Yes  Very High cost
    12  Yes           No   No     Medium cost
    13  Yes           No  Yes       High cost
    14  Yes           No  Yes        Low cost

    [15 rows x 21 columns]

    Total Number of Records : 2000
```

```
Column Names:
['Battery_power_mAh', 'Bluetooh', 'Speed_of_microprocessor', 'Dual_sim', 'Front_c
...

======================================================================
                    TARGET INFORMATION
======================================================================

Target Column : price_range

Target Values:
price_range
Medium cost       500
High cost         500
Very High cost    500
Low cost          500
Name: count, dtype: int64

Number of Input Features : 3

Input Features:
['Speed_of_microprocessor', 'Cores_of_processor', 'talk_time']

======================================================================
                    DATA SPLITTING
======================================================================

Training Records : 1400
Testing Records  : 600

======================================================================
               TRAINING RANDOM FOREST
======================================================================

Model Training Completed Successfully!


======================================================================
...                 ACTUAL VS PREDICTED
======================================================================
Record 1  | Actual Price Category = Low cost  | Predicted Price Category = Very High cost
Record 2  | Actual Price Category = High cost  | Predicted Price Category = Very High cost
Record 3  | Actual Price Category = Medium cost  | Predicted Price Category = High cost
Record 4  | Actual Price Category = Very High cost  | Predicted Price Category = High cost
Record 5  | Actual Price Category = Medium cost  | Predicted Price Category = Low cost
Record 6  | Actual Price Category = Medium cost  | Predicted Price Category = High cost
Record 7  | Actual Price Category = High cost  | Predicted Price Category = Very High cost
Record 8  | Actual Price Category = Low cost  | Predicted Price Category = Low cost
Record 9  | Actual Price Category = Very High cost  | Predicted Price Category = Very High cost
Record 10  | Actual Price Category = Medium cost  | Predicted Price Category = Very High cost
Record 11  | Actual Price Category = Low cost  | Predicted Price Category = High cost
Record 12  | Actual Price Category = Low cost  | Predicted Price Category = Medium cost
Record 13  | Actual Price Category = High cost  | Predicted Price Category = Medium cost
Record 14  | Actual Price Category = Very High cost  | Predicted Price Category = Medium cost
Record 15  | Actual Price Category = Very High cost  | Predicted Price Category = Low cost
Record 16  | Actual Price Category = High cost  | Predicted Price Category = Very High cost
Record 17  | Actual Price Category = Very High cost  | Predicted Price Category = Very High cost
Record 18  | Actual Price Category = Very High cost  | Predicted Price Category = High cost
Record 19  | Actual Price Category = Medium cost  | Predicted Price Category = High cost
Record 20  | Actual Price Category = Low cost  | Predicted Price Category = Very High cost
Record 21  | Actual Price Category = Low cost  | Predicted Price Category = Very High cost
Record 22  | Actual Price Category = High cost  | Predicted Price Category = Medium cost
Record 23  | Actual Price Category = Medium cost  | Predicted Price Category = Low cost
Record 24  | Actual Price Category = High cost  | Predicted Price Category = Low cost
Record 25  | Actual Price Category = Low cost  | Predicted Price Category = Medium cost
Record 26  | Actual Price Category = Medium cost  | Predicted Price Category = Low cost
Record 27  | Actual Price Category = Very High cost  | Predicted Price Category = Low cost
Record 28  | Actual Price Category = High cost  | Predicted Price Category = Low cost
Record 29  | Actual Price Category = High cost  | Predicted Price Category = Very High cost
Record 30  | Actual Price Category = Low cost  | Predicted Price Category = High cost
```

```
...  ======================================================================
                    MODEL PERFORMANCE
     ======================================================================

     Accuracy : 25.0 %

     Confusion Matrix:
     [[34 37 34 43]
      [42 32 45 32]
      [33 38 38 37]
      [25 36 48 46]]

     ======================================================================
                    FEATURE IMPORTANCE
     ======================================================================
                    Feature  Importance
                  talk_time    0.409530
     Speed_of_microprocessor    0.400538
          Cores_of_processor    0.189931

     ======================================================================
               MOBILE PRICE PREDICTION COMPLETED
     ======================================================================

     Target Column Used : price_range
     Number of Features : 3
     Training Samples   : 1400
     Testing Samples    : 600
     Model Used         : Random Forest Classifier
     Accuracy           : 25.0 %

     Final Result:
     The Random Forest algorithm successfully classified
     the mobile phones into their respective price categories.
     ======================================================================
```